

Lab 03 – GOOSE Spoof in a Lab Substation

Industrial Security (Master)

• Maps to Section 03

• 3 hours

LAB 03

🎯 Goal

Set up a minimal IEC 61850 lab on a single host, exchange GOOSE frames between two virtual IEDs, then spoof a higher-`stNum` frame and observe the subscriber's reaction. Conclude with the IEC 62351 mitigation.

✂ Step 1 – Build the lab

```
1 git clone https://github.com/mz-automation/libiec61850.git
2 cd libiec61850 && mkdir build && cd build
3 cmake .. && make -j
4 # Run the goose example publisher / subscriber on different interfaces
5 sudo ./examples/goose_publisher/goose_publisher_example eth0 &
6 sudo ./examples/goose_subscriber/goose_subscriber_example eth0 &
```

Capture with `tcpdump` or `Zeek`; confirm baseline GOOSE multicast.

✂ Step 2 – Capture and analyse

Open the PCAP in Wireshark with the IEC 61850 dissector enabled. Identify the GOOSE `appID`, `stNum`, `sqNum` fields. Note the highest `stNum` seen.

✂ Step 3 – Craft the spoof

Modify the publisher (or write a Scapy script) to emit a frame with `stNum` = `max+1`, signalling the opposite of the current binary state. Inject onto the same L2 segment.

✂ Step 4 – Observe and mitigate

Watch the subscriber log: it should immediately accept the higher state. Now enable IEC 62351-6 signed GOOSE in the publisher (`libiec61850` supports it via configuration). Re-inject; verify the subscriber rejects the unsigned spoof.

📁 Deliverable

PCAPs (baseline, spoof, signed-mitigation), screenshot of the subscriber log in each case, and a 1-page note covering: why `stNum` bypass works, why L2 isolation is the typical workaround in production substations, and what IEC 62351-6 fixes.

⚠ Caution

Only on an isolated VLAN with equipment you own. GOOSE in a real substation drives protection relays – a spoof can trip a feeder.